OmniAI Cloud*: An Intelligent Unified Multi-Modal AI System with Automatic Model Selection and Explainability*

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Abstract: Modern AI systems have to deal with different types of inputs, such as images, documents, and plain text, within a single system. However, most existing AI solutions are designed as separate pipelines, where users must manually select models and follow different workflows depending on the input type.

This paper proposes OmniAI Cloud, a ready-to-use multi-modal AI system that automatically detects the input type and selects suitable AI models using an intelligent routing agent. The system integrates computer vision, natural language processing, document analysis, multilingual translation, and explainability into a single platform.

The system activates only the required models and follows a modular design, making it more flexible and easier to deploy than traditional AI systems that are designed to perform a single task.

Keywords: Multi – Modal AI, Intelligent Agent, Automatic Model Selection, Explainable AI, Computer Vision, NLP, Document Intelligence

1. Introduction

Recent advances in deep learning have also enhanced the capabilities of various domains of artificial intelligence. In computer vision, deep residual learning allowed the training of very deep convolutional networks, resulting in significant improvements in image classification tasks [1]. In natural language processing, the transformer architecture introduced self-attention mechanisms that significantly enhanced scalability and parallel processing capabilities for sequence-based tasks [2]. Based on these concepts, BERT proposed bidirectional contextual representation learning and achieved state-of-the-art results in a wide range of NLP tasks [3].

However, most AI systems are still single-task systems and work independently. In practical scenarios, users tend to input different types of data, such as images, scanned documents, and multilingual text. Using multiple systems for different types of input data tends to increase complexity, hinder scalability, and hamper usability.

Another significant challenge is the lack of transparency in many AI systems. The predictions are made in a non-transparent

manner, which affects the trust of the users and the adoption of AI in real-world applications. The research in explainable artificial intelligence emphasizes the importance of interpretability and transparency in developing AI systems [4].

To overcome the above challenges, this paper proposes OmniAI Cloud, an intelligent, unified multi-modal AI system that automatically detects the input data, directs it to appropriate AI pipelines, and generates explainable outputs in a single production ready system.

Many of today's object detection algorithms, when trained on a large general dataset, have a tendency to classify animals with very general terms such as "bird" or "deer,"

which makes them less useful for wildlife research and other tasks that require species-level information. OmniAI Cloud addresses this problem by routing algorithms depending on the task and refining labels to the species level.

The all Deep Learning models in the OmniAI Cloud are used as pretrained models, without finetuning them. The contribution for this work is a system level : automatic routing, modular orchestration, label refinement, and explainability, evaluated on the curated multi modal test datasets.

The key contributions of this paper are:

- A unified three-layer architecture for multi-modal AI processing
- An intelligent auto-selector agent for dynamic model routing
- A smart label refinement mechanism that enhances the specificity of object detection
- A routing-level explainability framework
- A scalable and deployable system using state of the art AI models

2. Related Work

A. Computer Vision and Object Detection

Deep residual learning was another significant breakthrough in the field of computer vision, as it addressed the issues of degradation in deep neural networks [1]. The availability of large datasets, such as COCO, also helped push the frontiers of computer vision research by providing a common benchmark for object detection and instance segmentation tasks [5].

Single-stage object detection models, such as YOLO, enabled real-time processing, but since these models are trained on large datasets, they are known to output generic class labels with less information [6]. Vision language models, such as BLIP, enabled semantic understanding by linking visual concepts to natural language, enabling image captioning tasks [7].

Object detection and image captioning have been done well in previous work, but these models are generally developed and utilized as individual models. Compared to this, the OmniAI Cloud system integrates detection and classification models and directs each image to the appropriate model so that the results can be refined for each species, which is beneficial for images of wildlife.

However, it is a fact that despite all these breakthroughs, most of these models are still used independently and are not combined into a single system that can enable dynamic multi-modal processing.

B. Natural Language Processing

Transformer architectures subsequently replaced recurrent neural networks with self-attention mechanisms, greatly enhancing efficiency and scalability [2]. BERT pushed the state-of-the-art in contextual understanding of language with its bidirectional pre-training approach, achieving state-of-the-art results on a broad spectrum of NLP tasks [3]. DistilBERT proved that knowledge distillation can preserve most of BERT's performance with greatly reduced computational and memory complexity, making it amenable to practical applications [8]. The T5 framework integrated several NLP tasks into a text-to-text paradigm, allowing flexible summarization and text generation [9].

Large-scale multilingual translation models further extended the NLP paradigm by facilitating cross-lingual processing for a broad spectrum of languages [15].

C. Document Intelligence and OCR

OCR-based pipelines remain an integral component of document understanding systems. PaddleOCR is a robust deep learning-based approach for multilingual text detection and recognition [10]. Recent approaches combine OCR outputs with generative models to support automated document extraction and summarization [11]. OCR-free approaches such as Donut support end-to-end document understanding but require a large amount of training data and offer limited flexibility for heterogeneous workflows [12]. Layout-conscious and multimodal document intelligence approaches demonstrate the efficacy of integrating text, image, and structure knowledge for document-level reasoning [13].

D. Explainability and Unified AI Systems

Explainable AI methods like LIME stress the significance of transparency and trust in predictions made by the model [4]. Agent-based orchestration platforms demonstrate the strength of decision-driven model selection in complex multi-modal scenarios [14]. The above-mentioned studies illustrate the significance of holistic systems that not only combine multiple models of AI but also ensure transparency in routing and decision-making process.

Table: Comprehensive Comparison of Related work and OmniAI Cloud

Ref Paper Year	Area	Input Modality	Core Contribution	Auto Model Selection	Multi Model Orchestration	Explainability	Limitation vs OmniAI Cloud
[1] Deep Residual Learning	Vision	Image	Deep residual CNN for classification	No	No	No	Performs only single – task

(ResNet), 2016							
[5] Microsoft COCO Dataset, 2014	Vision	Image	Benchmark for Standardized detection	No	No	No	A dataset only not an application or system
[6] YOLO : Real Time Object Detection, 2016	Vision	Image	Fast and single stage of object detection	No	No	No	Only having Generic labels, no routing system
[7] BLIP : Vision Language Pretraining, 2022	Vision and Language	Image and Text	Captioning and VL alignment	No	No	No	Standalone VL model
[2] Attention Is All You Need (Transformer), 2017	NLP	Text	Self attention architecture model	No	No	No	Only an Architecture
[3] BERT, 2019	NLP	Text	Contextual language understanding	No	No	No	Task based specific model
[8] DistilBERT, 2019	NLP	Text	Lightweight transformer	No	No	No	A Single model deployment
[9] T5 (Text – to – Text Transformer), 2020	NLP	Text	Unified text generation framework	No	No	No	Not a multi modal system
[15] NLLB 200 Language MT, 2022	Translation	Text	Large scale multilingual MT	No	No	No	Language Translation only
[10] PP OCR (Paddle OCR), 2020	OCR	Image or PDF	Practical multilingual OCR	No	No	No	OCR only, no pipeline
[11] OCR + GenAI, 2023	Document AI	PDF or Image	OCR and Summarization	No	Partially	Partially	Not dynamic routing
[12] Donut (OCR Free Doc Understanding), 2022	Document AI	Document Image	End to End Document Transformer	No	No	No	Requires a specific task training
[13] Unified Multimodal Doc Framework, 2023	Document AI	Layout, Text and Image	An Enterprise level document intelligence	No	Partially	No	Heavy, but not modular
[4] LIME (Explainable AI), 2016	XAI	Anyone (Image, Text, Document etc)	Local Model explanation	No	No	Yes	Explains the predictions not routing system
[14] DocAgent, 2024	Agentic AI	Documents Multi Modal	Fully Agent Based orchestration	Partially	Yes	Partially	Focuses on only Documents
OmniAI Cloud (Proposed)	Unified Multi Modal AI Agent	Image, Text and PDF	Dynamic routing, modular AI and label refinement	Yes	Yes	Yes	Unified production ready system

Table 1: A brief comparison table of all reference papers

As we show in Table1, the prior works focus primarily on individual models or specific task pipeline systems. None of the reviewed systems provided the unified multi modal processing with in the dynamic routing, label refinement, and routing level explainability AI agent system within a single deployable architecture. OmniAI Cloud addresses these gaps by integrating pretrained models under an intelligent orchestration framework without addition model training.

3. System Architecture

OmniAI Cloud follows a three-layer architecture consisting of a user interface layer, an intelligence layer, and a model execution layer.

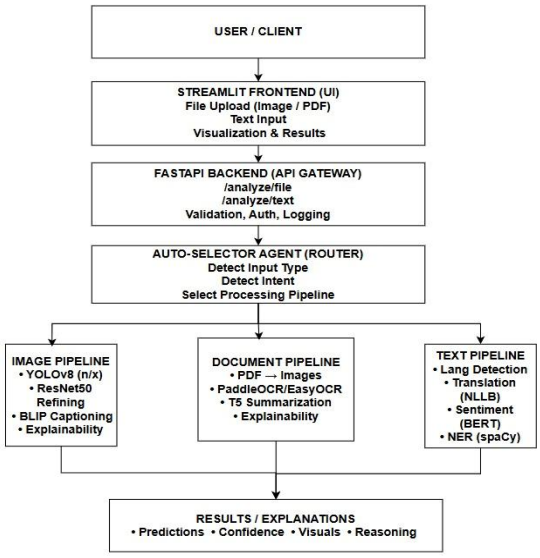

Fig 1: System Architecture of OmniAI

Figure depicts that the overview of the OmniAI Architecture flow that how it takes input and provides the output from user to frontend then after backend and auto model router model selection and then classifying them according to the category and activating the models and provides the results.

A. User Interface Layer

The user interface layer is comprised of a web application that has been developed using Streamlit, which allows users to upload images, documents, or text. The backend component of the application has been developed using FastAPI, which provides RESTful endpoints for handling analysis requests.

B. Intelligence Layer

The intelligence layer carries out the analysis of file signatures, metadata, and text features to identify the type of

input as image, document, or text. Depending on the analysis, an auto-selector agent automatically enables the corresponding AI pipelines.

To address the shortcomings of generic object detectors, a smart label refinement approach is employed, which integrates the predictions of image classification with object detection to improve semantic specificity.

The intelligence level of the system's design incorporates a routing technique that takes into consideration the presence of nearby wildlife when making routing decisions. If the uploaded image is first identified as containing an animal from a predetermined list of species, the system then changes from a small fast detection model to a larger wildlife model.

C. Model Execution Layer

The execution layer of the model integrates modular AI components, such as: ResNet-based image classification [1], YOLO-based object detection [6], PaddleOCR-based text extraction [10], BLIP image captioning [7], Transformer-based models for sentiment analysis and named entity recognition [3, 8], T5 text summarization [9], Multilingual translation models [15]. The models are managed through the use of lazy loading and the singleton pattern.

4. Methodology

A. Image Processing Pipeline

The images are processed through deep residual networks to extract semantic context [1]. Object detection is carried out through YOLO architectures [6]. When text content is detected, the OCR process is initiated through PaddleOCR [10]. Vision-language captioning is applied through BLIP to produce semantic image descriptions [7].

In images containing animals, the two-step process is followed by OmniAI Cloud. We begin by classifying the image to reduce the list of possible species, and the result is used to select the best model for the task. If the classified category belongs to a predefined list of wildlife species, then the model changes to a higher-capacity detection model.

B. Text Processing Pipeline

Language detection is carried out as a preprocessing task, followed by multilingual translation using larger-scale translation models if needed [15]. Sentiment analysis and named entity recognition are carried out using transformer encoders [3, 8]. Text summarization is performed using the T5 model [9].

C. Document Processing Pipeline

Text-based PDF files are processed directly, whereas scanned files are converted to image and processed using OCR [10]. The processed text is then formatted and summarized using NLP models, which allows for automated document understanding [11]

D. Explainability Module

The explainability module provides human understandable explanations of input classifications, active models, processing operations, and final results. This approach enhances explainability and follows established principles of explainable artificial intelligence [4].

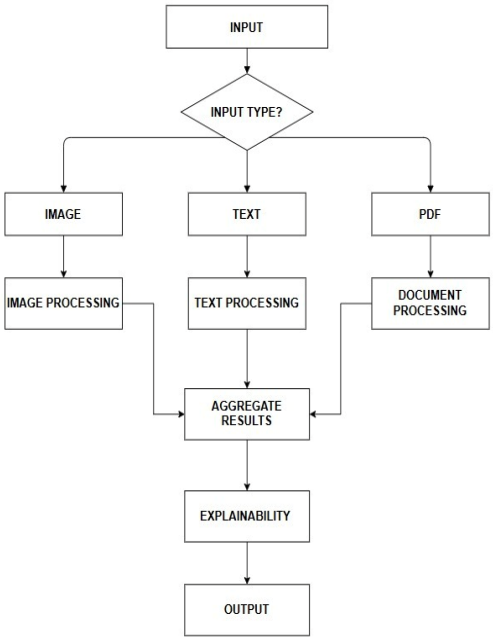

Fig 2: Work Flow Diagram

Figure 2 presents the workflow of the system from user interaction to display the explainable output

5. Implementation Details

The system is developed in Python with FastAPI for backend development and Streamlit for frontend development. Deep learning modules are developed using PyTorch, Vision modules (ResNet, YOLO, BLIP) and Hugging Face Transformers. The system is developed in such a way that it can be executed on CPU and has the option of using GPU. The system uses lazy loading and caching for reduce memory overhead and late start. For improving the production readiness, pre – initialization is applied for heavy components such as OCR and translation models, that enables faster first response times.

6. Results and Discussion

All experiments are used in these system are publicly available pretrained models. No additional training or fine tuning was performed. The improvements observed are due to dynamic routing, automatic model selection, modular orchestration, and label refinement strategies rather than model training.

6.1. Wildlife Accuracy and Label Refinement

OmniAI system improves the wildlife recognition accuracy by combining routing with label refinement. As shown in Fig. 3, the system achieves 95% wildlife detection accuracy compared to 78% (YOLOv8n) and 82% (ResNet50). This improvement is achieved trough multi model routing instead of training new models.

Comparison table

Method	Model Strategy	Species Specificity	Accuracy
YOLOv8n	It is a single detection model	It was supposed to specify the Generic labels	78%
ResNet50	It is used for Classification Purpose only	It was supposed to specify the Image level class	82%
YOLOv8x	It is used for the High capacity detection	It was supposed to specify partially	91%
Ensemble(No refinement done)	It used for the purpose both Detection and Classification	It was supposed for the Limited refinements	89%
OmniAI Cloud	It was completely used for purpose of Routing and label refinements	**132 species**	**95%**

Table 2: Comparison Table for different methods with OmniAI

Table 2 presents the comparison of different methods for detection and classification with the proposed OmniAI Cloud approach. The methods, such as YOLOv8n and YOLOv8x use a single model for the purpose of object detection, leading to the classification results being generally or partially refined only in some cases. ResNet50 is used at the image level to classify the entire image, rather than the specific image objects or features within the image. An average approach using a basic ensemble shows slightly better results, but they are average only because the approach is not supported by any level of intelligent refinement and only shows weaker classification accuracy compared to the OmniAI Cloud approach, which successfully identified 132 animal species with the highest level of accuracy.

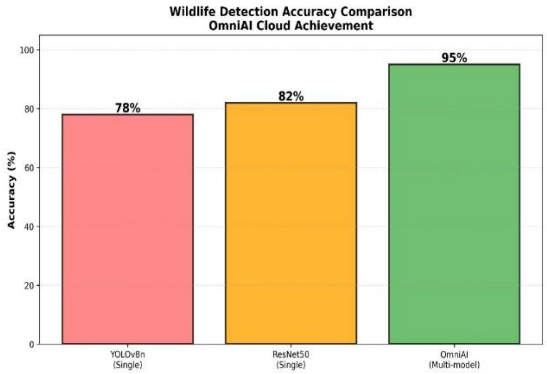

Fig 3: Accuracy comparison with YOLOv8n vs ResNet50 vs OmniAI

The above figure indicates the comparison of the accuracy of the detection of wildlife by the respective methods. As shown, the use of YOLOv8n as a standalone approach can give an accuracy of up to 78 percent, mainly because of the generic outputs of object detection. Using the ResNet50 model shows that it can increase the overall model's accuracy to 82 percent by implementing the classification approach, though the model ends up being less accurate in the detection of objects. The results of OmniAI Cloud show that it can achieve the highest, which is 95 percent, by implementing the intelligent routing approach, which involves the refinement of the labels.

6.2. Species Coverage Expansion

A best grained coverage increases from 38 species (old systems) to 132 species (OmniAI Cloud) using the both routing and refinement process, as shown in Fig.4. This improves usability in wild life classification scenarios and best for detection of wildlife's where the generic labels are insufficient.

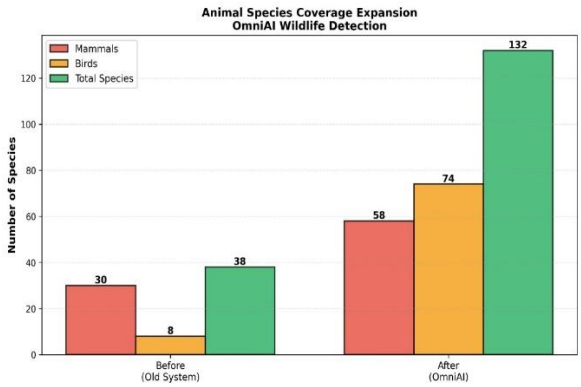

Fig 4: Comparison of the Animal Coverage Expansions i.e.., Old System vs OmniAI

Figure shows the expansion of the animal species coverage for the OmniAI wildlife system. As seen from the figure, initially the system was only capable of detecting 38 animal species in total, 30 of which were mammals, while 8 were birds. When using the new OmniAI cloud, it expanded to 132 animal species in total, with 58 of them being mammals while 74 were birds.

6.3. What happens if we take some models out

Variant	Auto Model Selector	Label Refinement	Pre Initialization	Accuracy	First Request Latency (sec)
Full OmniAI	Keep	Keep	Keep	95%	1.8 to 2.5 sec
Without Auto Selector	Remove (all models run at a time)	Keep	Keep	95%	12.4 to 15 sec
Without Label Refinement	Keep	Remove	Keep	86%	2 sec
Without Pre Initialization	Keep	Keep	Remove	95%	84 to 97 sec

The comparison shows the effects of different modules contained within OmniAI Cloud. If the auto model selector module is not included, so that all models operate simultaneously, response accuracy remains unchanged while response time increases substantially. However, if the label refinement module is commented out, response accuracy is negatively impacted, thus demonstrating the importance of the module to effective species identification. As expected, disabling the pre-initialization feature fails to change the response accuracy, but produces a very large increase in the first request latency.

6.4. Limitations

We had built OmniAI Cloud using powerful pretrained models like BLIP, PaddleOCR, but we know that not every environment or system has unlimited power. To stay flexible, we are developing lighter model variants that fits into tighter spaces without losing their core edge. We also noticed that the quality of scan matters, if a document is distorted, our OCR (text extraction) has some hard time to get 100% accurate results. To fix this, our further steps to involve adding automatic document cleanup and 'smart traffic controller' (the best routing strategy that to assign each job) to pick the best model for every job.

7. Conclusion and Future Works

Finally in this paper, we introduce OmniAI Cloud – a versatile AI hub built to handle a little bit of everything. Instead of you manually telling the system what it's looking at, which model that system wants to use, our OmniAI identifies the input types automatically, and selects the best model by using unified "brain" for the job, and coordinates all the moving parts behind the scenes. We had even made sure that the system can 'explain' why it choose that specific model and the path, so it is not just a mysterious black box.

Our test on specialized data show that this setup is particularly good at identifying wildlife accurately and getting up to show the speed quickly (cold start performance) without needing a large infrastructure. In future, we are planning to teach the system how to make even run smarter routing decision system by its own, and we will about to upgrade by including the training of the datasets and expand the skills to Video and Audio analysis, and run even bigger stress test across multiple platforms and industries to see just how far it can go.

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